

SUNY Polytechnic Institute

DSA 506: Visual Analytics and Communications

Project 1: Exploratory Data Analysis and Dashboard Development

Due Date	See Brightspace for submission deadline
Team Size	Individual submission
Deliverables	Dashboard (deployed on streamlit cloud)), video presentation up to 5 mins, code and written summary (2-3 pages max)

This project requires selecting a real-world dataset, conducting a structured exploratory data analysis following the QUEST framework, building an interactive dashboard to communicate findings, and delivering a video presentation that tells a compelling data story to a non-technical audience. The emphasis throughout is on analytical reasoning and visual communication rather than code demonstration. No code should appear in any deliverable presented to the audience.

Your analysis should follow the QUEST framework, a five-phase methodology for structured exploratory data analysis that synthesizes six decades of EDA scholarship. The framework rests on three guiding principles: EDA is detective work, not bookkeeping, meaning every visualization should answer a specific question and every answer should generate further questions; the quality of exploration depends on the disposition of the explorer, including the willingness to be surprised and the discipline to document the investigative journey; and automation accelerates but does not replace human judgment.

Deliverables

1. Interactive Dashboard

Build and deploy an interactive dashboard using Streamlit (or an equivalent tool). The dashboard should satisfy the following requirements.

- Present at least four distinct visualizations that address your analytical questions.
- Include at least one interactive element that allows the user to filter, select, or adjust what is displayed.
- Provide brief contextual text within the dashboard that orients the user to the data, the questions being addressed, and how to interpret the visualizations.
- Deploy the dashboard to a publicly accessible and verify that it loads correctly before submission.

- e. Include a section or page within the dashboard that documents the data source, the date it was accessed, and how the data could be updated in the future to keep the dashboard current.

2. Video Presentation

Record a video presentation of no more than five minutes. The video should be structured for the data's target audience, not for an academic or technical audience. It should cover the following elements.

- a. The problem or decision context that motivates the analysis.
- b. A walkthrough of the key findings, using the dashboard or selected visualizations as supporting material.
- c. The storytelling arc connecting the original questions to the evidence and conclusions.
- d. Actionable insights or recommendations derived from the analysis.
- e. No code, no technical jargon unexplained to the audience, and no screen recordings of development environments. The presentation should feel like a briefing or a data journalism segment, not a code walkthrough.

3. Written Summary

Submit a brief written document (two to three pages) that complements the dashboard and video. This summary should include the following.

- a. A description of the dataset, including source, size, key variables, and any preprocessing performed.
- b. The analytical questions posed in Phase 1 and how they evolved during the project.
- c. A brief account of the QUEST phases as applied in your project.
- d. A reflection on what the data revealed, what limitations you encountered, and what further analysis could be pursued.

Data Requirements

Use a real, publicly available dataset. Simulated or toy datasets (such as Iris, Titanic, or mtcars) are not permitted. The dataset should be rich enough to support meaningful exploration across multiple variables and should connect to a genuine problem or decision context. Examples of appropriate sources include government open data portals, Kaggle datasets with real-world provenance, publicly available APIs, organizational data shared with permission, and curated research repositories.

Document the following information about their data source in both the dashboard and the written summary: the original source and URL, the date of access or download, the license or terms of use, and a description of how the data could be refreshed or updated in the future. This documentation supports the principle of data sustainability, ensuring that the analytical product you build can remain a living tool rather than a static artifact.

Evaluation Criteria

Projects will be evaluated according to the following criteria and weights.

Criterion	Weight	Deliverable
Analytical depth and adherence to the QUEST framework	25%	Summary
Visualization quality, clarity, and appropriateness of chart selection	25%	Dashboard
Dashboard functionality, interactivity, and deployment	20%	Dashboard
Storytelling effectiveness and audience awareness	20%	Video
Data documentation and sustainability	10%	Summary / Dashboard